

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T28 · Forming and Solving Equations

With Answers

Questions with full worked solutions

■ 5 pages

FAMILY MAP

Forming Solving Equations

F01

Form and solve a quadratic from data

2 Q | 9 marks

- 5 Yasmin has some identical rectangular tiles.
Each tile is L cm by W cm.

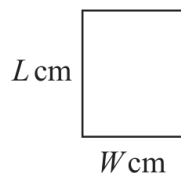


Diagram **NOT**
accurately drawn

Using 9 of her tiles, Yasmin makes rectangle $ABCD$, shown in the diagram below.

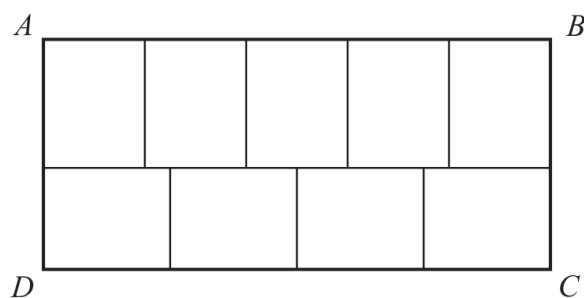


Diagram **NOT**
accurately drawn

The area of $ABCD$ is 1620 cm^2

Work out the value of L and the value of W .

$L = \dots\dots\dots$ $W = \dots\dots\dots$

Worked solution

November 2021 | Question 5 | 5 marks

Family: Form and solve a quadratic from data | Variant: Area or dimension quadratic model

METHOD / WORKING

1. The top row has 5 tile widths and the bottom row has 4 tile lengths, so

2. $5W=4L$

3. $L=\frac{5}{4}W$

4. The rectangle has width $5W$ and height $L + W$:

5. $5W(L+W)=1620$

6. Substitute $L = \frac{5}{4}W$:

7. $5W\left(\frac{5}{4}W+W\right)=1620$

8. $5W\left(\frac{9}{4}W\right)=1620$

9. $\frac{45}{4}W^2 = 1620$

10. $W^2 = 144$

11. $W=12, L=\frac{5}{4}(12)=15$

12. Answer: $L = 15\text{cm}, W = 12\text{cm}$.

FINAL ANSWER

L=15 cm, W=12 cm.

- 2 Here are two rectangles, rectangle A and rectangle B .

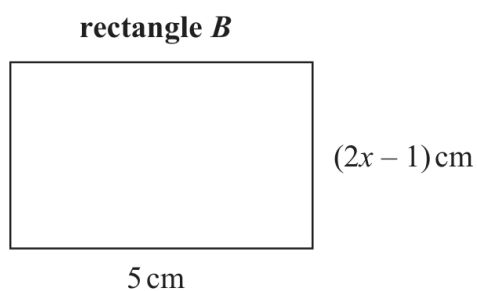
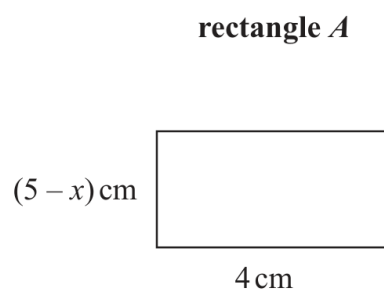


Diagram **NOT**
accurately drawn

The area of rectangle B is twice the area of rectangle A .

Work out the value of x .

Show your working clearly.

$x = \dots\dots\dots$

Worked solution

November 2021 | Question 2 | 4 marks

Family: Form and solve a quadratic from data | Variant: Area or dimension quadratic model

METHOD / WORKING

1. Area of rectangle A:

2. $4(5-x)$

3. Area of rectangle B:

4. $5(2x-1)$

5. Rectangle B has twice the area of rectangle A:

6. $5(2x - 1) = 2 \cdot 4(5 - x)$

7. $10x-5=40-8x$

8. $18x=45$

9. $x=2.5$

10. Answer: $x = 2.5$.

FINAL ANSWER

$x=2.5.$